

Juice Uses Earth's Gravity to Help It Reach Jupiter

From: ScienceDaily All Mined: 2026-10-04 5 min read

The European Space Agency's Jupiter Icy Moons Explorer, known as Juice, passed Earth on September 28, 2026. The spacecraft used Earth's gravity to change its route toward Jupiter while using very little fuel.

This kind of move is called a **gravity assist**. A planet's gravity can help change a spacecraft's speed and direction. During this **flyby**, Juice's direction changed by 20 degrees, and its speed increased by 3.5 kilometres per second.

At its closest point, Juice was 8,640 kilometres above the Indian Ocean. Its monitoring cameras took a series of pictures during the encounter.

Reaching the right path required careful planning. In the four weeks before the **flyby**, mission controllers made just one small **course correction**. They had set aside six possible opportunities to make adjustments.

Angela Dietz, Juice's spacecraft operations manager, said the team's planning helped it use only a small amount of the fuel reserved for the **flyby**. Saving that fuel leaves more available for observations of Jupiter's icy moons.

The Earth visit also gave scientists a chance to check Juice's instruments. **Calibration** means checking and adjusting instruments so their measurements can be used reliably. Earth provided a familiar planet for these tests.

Teams spent months planning activities with the groups responsible for Juice's 10 scientific instruments. They had to decide how to use the spacecraft's available time, power and other resources.

Juice project scientist Claire Vallat explained that the journey offers only a few chances to test the instruments under well-understood conditions. When Juice passed through Earth's shadow, it relied on battery power alone. The teams gave priority to tests that would help prepare the instruments for Jupiter.

The spacecraft also spent several days travelling through Earth's **magnetotail**. This is the long part of Earth's magnetic field that stretches away from the Sun.

Juice is scheduled to make 35 **flybys** of Jupiter's moons Ganymede, Callisto and Europa. Planning for those encounters began years ago. Its Europa visits are expected in the early 2030s.

The Earth **flyby** served as a rehearsal, helping scientists learn how the instruments work in space and prepare to study future measurements. Controllers are continuing their close tracking of Juice through October 10.

Word Treasure

gravity assist -- Using a planet's gravity to change a spacecraft's speed and direction.

course correction -- An adjustment to a spacecraft's path.

calibration -- Checking and adjusting instruments so their measurements can be used reliably.

magnetotail -- The long part of Earth's magnetic field that stretches away from the Sun.

flyby -- An encounter in which a spacecraft passes a planet or moon.

Background you need

A planet's gravity can change a spacecraft's speed and direction during a gravity assist. Earth provided a familiar planet for checking and adjusting Juice's instruments. Earth's magnetotail is the part of its magnetic field that stretches away from the Sun.

Break it down

Main event: Juice uses Earth's gravity to change its route toward Jupiter.

Encounter details: Changes in direction and speed, closest altitude and camera observations are described.

Mission planning: Course corrections and fuel savings show how controllers prepared for the flyby.

Instrument preparation: Calibration, resource planning and battery use explain the scientific tests.

Future mission: Planned moon flybys and continued tracking connect the Earth rehearsal to later observations.

Why it matters

The flyby conserved fuel for observations of Jupiter's icy moons and gave scientists a chance to test instruments under familiar conditions before future encounters.

Test what you remember

1. How did Juice's speed change during the Earth flyby?
 - (A) It increased by 3.5 kilometres per second.
 - (B) It decreased by 3.5 kilometres per second.
 - (C) It stayed the same.
 - (D) It increased by 20 kilometres per second.
2. Where was Juice at its closest point to Earth?
 - (A) 20 kilometres above Ganymede
 - (B) 8,640 kilometres above the Indian Ocean
 - (C) 3.5 kilometres above the Indian Ocean
 - (D) 8,640 kilometres above Europa
3. How many possible adjustment opportunities had controllers set aside before the flyby?
 - (A) Six
 - (B) Four
 - (C) Ten
 - (D) One
4. According to Angela Dietz, what was a benefit of saving fuel during the flyby?
 - (A) It left more fuel available for observations of Jupiter's icy moons.
 - (B) It supplied power in Earth's shadow.
 - (C) It removed the need for instrument tests.
 - (D) It ended the need to track Juice.
5. Which tests did teams give priority to during the Earth encounter?
 - (A) Tests that would cancel the Europa visits
 - (B) Tests that would stop Juice from entering the magnetotail
 - (C) Tests that would replace all 10 instruments
 - (D) Tests that would help prepare the instruments for Jupiter
6. When are Juice's Europa visits expected?
 - (A) On October 10
 - (B) In the early 2030s
 - (C) On September 28, 2026
 - (D) During the four weeks before the Earth flyby

Answer key (try the quiz first!): 1: A 2: B 3: A 4: A 5: D 6: B

Questions to think about

1. **Angela Dietz, Juice's spacecraft operations manager:** She credited the team's planning with using only a small amount of the fuel reserved for the flyby.
2. **Claire Vallat, Juice project scientist:** She explained that the journey provides only a few opportunities to test instruments under well-understood conditions.

MY THOUGHTS (at least 20 words):